COMPUTATIONAL COUPLING — Comprehensive Literature Review

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> **Status: planning document, superseded.** This was written at the start of the project (September 2026), before any experiment was run, and it records the research plan and novelty analysis as they stood then. It is **not** a statement of findings. Several of its premises did not survive: the human H01 microscale tier was withdrawn when its connectivity graph turned out to be empty, the CoupledEventNet / neuromorphic-efficiency direction was withdrawn along with the fabricated silicon figures supporting it, and the central motif hypothesis it sets out was tested and came out with the opposite sign.
>
> For what the project actually found, see `../info.md` and the manuscript. For what was withdrawn and why, see `../AUDIT.md`. This file is kept because the retraction is only checkable against the plan it replaced.

Project: COMPUTATIONAL COUPLING
Lead Autonomous Research Engineer: Antigravity
Version: 1.0 (Milestone 0)
Date: September 17, 2026

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1. Introduction: From Static Connectomics to Dynamic Event-Driven Computation

Over the past decade, high-throughput serial section electron microscopy (EM), automated computer vision segmentation, and massive community proofreading have delivered the first dense wiring diagrams of complex nervous systems:

1. The sexually dimorphic and developmental connectomes of *Caenorhabditis elegans* (Cook et al., 2019; Witvliet et al., 2021).
2. The whole-brain adult connectome of *Drosophila melanogaster* (FlyWire Consortium; Dorkenwald et al., 2024; Schlegel et al., 2024; Lin et al., 2024).
3. The millimeter-scale mammalian cortical volume of *Mus musculus* (MICrONS Consortium, 2025; Schneider-Mizell et al., 2025).

A central goal across systems neuroscience and machine learning has been to understand what computational advantages biological wiring confers over conventional artificial neural networks. However, current research is polarized into two disconnected regimes:

- **Descriptive Graph Connectomics:** Treating connectomes as static topological graphs, cataloging degree distributions, modularity, small-worldness, and 3-node subgraph frequencies (Milo et al., 2002; Sporns & Kötter, 2004; Varshney et al., 2011; Lin et al., 2024). These studies show that certain subgraphs are statistically overrepresented, but leave unanswered whether and how these structures perform dynamic computation.
- **Biomimetic Machine Learning & Reservoirs:** Constraining artificial recurrent networks or reservoirs with empirical connectomes (Hasani et al., 2020; Lappalainen et al., 2024; Suárez et al., 2024). In these approaches, the biological wiring is either hand-abstracted (NCPs) or frozen as a rigid tensor mask, ignoring biological remodeling, structural degeneracy, and asynchronous event-driven execution.

This review synthesizes the empirical datasets, computational models, and prior art across our three model organisms, formalizing the exact scientific gap addressed by **COMPUTATIONAL COUPLING**.

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2. Organism Systems & Empirical Datasets

System A — *Caenorhabditis elegans*

Caenorhabditis elegans possesses 302 somatic neurons in the adult hermaphrodite and 385 neurons in the adult male.

- **Anatomical Datasets:** The foundation laid by White et al. (1986) was modernized by Varshney et al. (2011) and fully completed by Cook et al. (2019), who mapped both sexes, distinguishing chemical synapses (directed) and gap junctions (undirected electrical couplings). Witvliet et al. (2021) reconstructed eight isogenic worms across post-embryonic development (L1 to adult), revealing that while the core scaffold is strictly stereotyped, peripheral connections undergo continuous remodeling and synaptic pruning. Ripoll-Sánchez et al. (2023) mapped the "wireless" neuropeptidergic and monoaminergic connectomes, demonstrating multi-layered neuromodulatory control.
- **Biophysical & Computational Infrastructure:** The OpenWorm project, `ConnectomeToolbox`, and `CElegansNeuroML` provide unified programmatic interfaces to these datasets, including neurotransmitter identities (GABA, acetylcholine, glutamate, dopamine, serotonin) and cell classes (sensory, interneuron, command, motor).

System B — *Drosophila melanogaster* (FlyWire Whole-Brain)

The adult female *Drosophila melanogaster* brain contains 139,255 neurons and over 50 million synapses.

- **FlyWire Reconstructions:** Published in *Nature* (October 2024), Dorkenwald et al. (2024) and Schlegel et al. (2024) generated the first complete whole-brain wiring diagram (v630 snapshot). Over 8,453 consensus cell types were categorized, and machine learning models predicted primary neurotransmitters (acetylcholine [Ach+], glutamate [Glu-], GABA [GABA-], dopamine, octopamine, serotonin).
- **Network Statistics & Modeling:** Lin, Yang et al. (2024) analyzed network statistics, revealing a heavy-tailed degree distribution, strong reciprocity in central brain regions, and an extensive rich-club forming a communication backbone. Shiu et al. (2024) built a full-brain Leaky Integrate-and-Fire (LIF) model (`Drosophila_brain_model`), proving that synapse-level connectivity and sign predictions alone suffice to replicate behavioral reflexes (e.g. proboscis extension, antennal grooming). Concurrently, Lappalainen et al. (2024) showed that constraining visual recurrent networks with optic lobe connectivity discovers biological motion detection mechanisms.

System C — *Mus musculus* (MICrONS Visual Cortex Circuit)

Rather than an unfeasible whole-brain reconstruction, mammalian validation relies on the Machine Intelligence from Cortical Networks (MICrONS) millimeter-scale cubic volume in the mouse primary visual cortex (V1) and higher visual areas (AL, LM, RL).

- **Multi-Modal Dataset:** MICrONS Consortium (2025) released dense EM reconstructions (~100,000 cells, ~500M synapses) co-registered with *in vivo* two-photon calcium imaging of neuronal responses to natural movies and directional drift gratings.
- **Inhibitory Motif Organization:** Schneider-Mizell et al. (2025) uncovered structured "motif groups" among inhibitory interneurons: parvalbumin (PV), somatostatin (SST), and vasoactive intestinal peptide (VIP) interneurons coordinate perisomatic and dendritic inhibition onto pyramidal targets, with VIP interneurons forming disinhibitory circuits that gate basket cell activity.

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3. Conserved Network Motifs Across Scales

Network motifs are defined as subgraphs that occur at significantly higher frequencies than expected in degree-preserving randomized null graphs (Milo et al., 2002). In comparative graph analyses, Lin et al. (2024) demonstrated that across *Drosophila*, *C. elegans*, zebrafish, and mouse Pinky100:

1. **Feed-Forward Loops (FFLs):** A 3-node motif where neuron A projects to B and C, and B projects to C. Coherent FFLs (where direct and indirect paths share sign) act as sign-sensitive delays or persistence filters; incoherent FFLs act as pulse generators and fold-change detectors (Alon, 2007).
2. **Reciprocal Feedback (RFB) & Mutual Inhibition:** Strong reciprocal connectivity (especially inhibitory-inhibitory or excitatory-inhibitory) is heavily enriched in decision and central integration neuropils across all three organisms.
3. **Disinhibitory Motifs:** An inhibitory neuron targeting another inhibitory interneuron to release an excitatory target from suppression, conserved from nematode locomotion switches (AVA/AVB) to mammalian cortical gating (VIP $\rightarrow$ PV $\rightarrow$ Pyramidal).

Critical Scientific Observation: Lin et al. (2024) established the topological overrepresentation of these motifs, but performed no functional, event-driven, or computational benchmarks.

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4. Prior Work in Biologically Inspired AI Architectures

Neural Circuit Policies (NCPs) & Liquid Time-Constant (LTC) Networks

Lechner, Hasani et al. (2020, 2021) showed that abstracting the 4-layer *C. elegans* sensorimotor structure into continuous-time ODEs yielded models with superior parameter efficiency (19 neurons vs. thousands in standard RNNs), interpretability, and out-of-distribution robustness on autonomous driving tasks. However, NCPs rely on hand-crafted simplified graphs, do not leverage empirical connectomes from flies or mammals, and require dense numerical ODE integration rather than sparse event-driven spikes.

Connectome-Informed Reservoir Computing (`conn2res`)

Suárez, Misic et al. (2024) introduced `conn2res`, embedding empirical structural connectomes (human diffusion MRI, mouse tract-tracing, *C. elegans*) into fixed recurrent reservoirs and training linear readouts on cognitive tasks. They found that biological connectomes operate near the edge of chaos, balancing dynamic memory and non-linear expressivity. However, `conn2res` treats the network topology as entirely immutable, lacks asynchronous spiking dynamics, and does not investigate structural remodeling.

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5. Neural Degeneracy & Biologically Constrained Remodeling

In biological nervous systems, structural degeneracy—the ability of structurally disparate neural circuits to perform indistinguishable dynamical functions—is a universal principle (Edelman & Gally, 2001; Prinz, Bucher, & Marder, 2004; Marder & Taylor, 2011). In the stomatogastric ganglion of crustaceans, 3-neuron circuits with 3- to 5-fold variations in synaptic and membrane conductances produce identical triphasic pyloric rhythms.

In machine learning, algorithms such as **Deep Rewiring (DEEP R)** (Bellec et al., 2018) dynamically prune and generate synapses during training to enforce extreme sparsity. However, DEEP R rewires artificial networks using gradient heuristics without biological validity: it ignores Dale's principle (violating the strict segregation of excitatory and inhibitory neurons), disregards neurotransmitter identities, and does not preserve biological motif invariants.

The Unexplored Opportunity:
By combining biological constraints (Dale's E/I sign preservation, neurotransmitter identity, bounded delays, degree distributions) with multi-objective optimization, we can quantitatively map the **biological degeneracy frontier**: measuring exactly how much an empirical connectome can be structurally deformed (Δ_{struct}) while maintaining dynamic task performance and minimizing spike count (energy proxy).

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6. Event-Driven Computing & Neuromorphic Efficiency

Modern deep learning architectures (Transformers, CNNs) are energy-intensive because they perform dense matrix multiplications on synchronous clocks. Neuromorphic architectures (e.g. Intel Loihi; Davies et al., 2018) operate on an asynchronous, event-driven paradigm where compute occurs only when a neuron emits a discrete spike.

Energy consumption in neuromorphic systems is directly proportional to the number of **Synaptic Operations (SynOps)**:

$$\text{SynOps} = \sum_{t=1}^{T} \sum_{i \in \text{spiking}(t)} k_{\text{out}}(i)$$

where $k_{\text{out}}(i)$ is the fan-out of neuron i.

Because biological connectomes have evolved under intense metabolic pressure, their modular motif structure (sparse feed-forward pathways, local reciprocal inhibition) may represent nature's optimized event-driven routing protocol.

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7. Conclusion: The COMPUTATIONAL COUPLING Opportunity

Existing literature has mapped the anatomy (Lin, Cook, MICrONS), simulated whole brains rigidly (Shiu), or abstracted toy nematode topologies into continuous ODEs (Hasani).

COMPUTATIONAL COUPLING occupies the rigorous, unoccupied intersection:

1. Extracting dynamic event-driven functional properties from conserved motifs across *C. elegans*, *Drosophila*, and *Mus musculus*.
2. Remodeling empirical circuits under biological constraints to trace the Pareto degeneracy frontier.
3. Discovering which motifs are non-negotiable computational invariants versus disposable biological scaffolding.
4. Synthesizing these invariants into an event-driven neural computing architecture that provides measurable efficiency and latency gains over conventional SNNs and reservoir baselines.